

Solution Key to Problem Sheet 1

Ans 1:

1.a $\sin(\omega t)$ gives $\omega_1 = \frac{2}{3}, f_1 = \frac{1}{3\pi}$

And Similarly, $\omega_2 = \frac{2\pi}{3}, f_2 = \frac{1}{3}$

Fundamental frequency (f_0): GCD (f_1, f_2)

However, GCD ($\frac{1}{3\pi}, \frac{1}{3}$) does not exist as π is an irrational number. So, $x(t)$ is an **Aperiodic signal**.

1.b $E_v \{ \sin(4\pi t) u(t) \} = \frac{[\sin(4\pi t)u(t) - \sin(4\pi t)u(-t)]}{2}$

Which is **Aperiodic signal**.

1.c It is a **Periodic signal**

$$\omega_0 = 0.8\pi;$$

$$N = \frac{2\pi k}{\omega_0} = \frac{5k}{2}. \text{ Thus, the fundamental period is 5.}$$

1.d Aperiodic

$$\omega_0 = 4; \frac{2\pi}{\omega_0} = \frac{2\pi}{4} \text{ is irrational.}$$

1.e Periodic

$$\omega_1 = \frac{\pi}{3}, N_1 = 8$$

$$\omega_2 = \frac{\pi}{8}, N_2 = 16$$

$$\omega_3 = \frac{\pi}{2}, N_3 = 4$$

$$\text{Fundamental period, } N_0 = \text{LCM}(N_1, N_2, N_3) = 16$$

1.f Periodic

With period 4

Ans 2:

2.a $x(n-3)$ is time delayed signal and shifted by 3.

It is zero for $n < 1$ and $n > 7$

2.b $x(n+4)$ is time advanced signal

It is zero for $n < -6, n > 0$

2.c Zero for $n < -4, n > 2$

2.d Zero for $n > 4, n < -2$

2.e Zero for $n < -6, n > 0$

Ans 3:

a) $x[2n] = \{6, 2, 7, 3\}$

b) $x[\frac{n}{3}] = \{\dots, 6, 0, 0, 3, 0, 0, 2, 0, 0, 5, 0, 0, 7, 0, 0, 9, \dots\}$

Ans 4:

a) $x(t)$ is periodic with period T ; $y_1(t)$ is periodic with $T/2$

b) $y_1(t)$ is periodic with period T ; $x(t)$ is periodic with $2T$

c) $x(t)$ is periodic with period T ; $y_2(t)$ is periodic with $2T$

d) $y_2(t)$ is periodic with period T ; $x(t)$ is periodic with $T/2$

Detailed solution for part (a):

$x(t)$ is periodic $\Rightarrow x(t + T) = x(t)$

Now $y(t + P) = x(2t + 2P) = x(2t)$, if $2P = kT$, where $k \in \mathbb{Z}$.

$\Rightarrow P = (kT/2)$. The fundamental period is obtained for $k = 1$ which is $T/2$. Likewise for parts (b), (c), (d)

Ans 5:

5.1) True. if $x[n + N] = x[n]$ then $y_1[n + N_0] = y_1[n]$ where $N_0 = \frac{N}{2}$ if n is even and $N_0 = N$ if n is odd.

5.2) False. $y_1[n]$ being periodic does not imply $x[n]$ is periodic.

Let $x[n] = g[n] + h[n]$ where

$$g[n] = \begin{cases} 1, & n \text{ even} \\ 0, & n \text{ odd} \end{cases} \quad \text{and} \quad h[n] = \begin{cases} 0, & n \text{ even} \\ (1/2)^n, & n \text{ odd} \end{cases}$$

Then $y_1[n] = x[2n]$ is periodic but $x[n]$ is clearly not.

5.3) True. if $x[n + N] = x[n]$ then $y_2[n + N_0] = y_2[n]$, where $N_0 = 2N$

5.4) True. if $y_2[n + N] = y_2[n]$ then $x[n + N_0] = x[n]$, where $N_0 = N/2$

Ans 6:

6. a) Consider $\sum_{n=-\infty}^{\infty} x[n] = x[0] + \sum_{n=1}^{\infty} (x[n] + x[-n])$

For odd functions, $x[n] + x[-n] = 0$, $x[0] = 0$

So, $\sum_{n=-\infty}^{\infty} x[n] = 0$

6. b) Let $y[n] = x_1[n] \cdot x_2[n]$

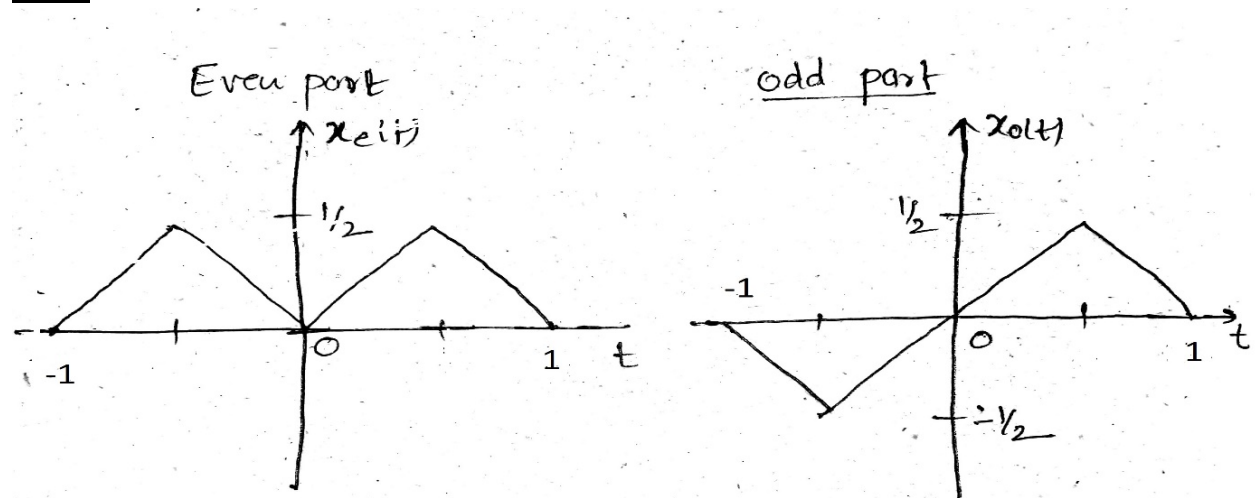
Then $y[-n] = x_1[-n] \cdot x_2[-n] = -x_1[n] \cdot x_2[n] = -y[n]$

Thus $y[n]$ is an odd signal.

6. c) $\sum_{n=-\infty}^{\infty} x^2[n] = \sum_{n=-\infty}^{\infty} (x_e[n] + x_o[n])^2$

Use $\sum_{n=-\infty}^{\infty} (x_e[n] x_o[n]) = 0$ to obtain the result.

Ans 7:



Ans 8:

a) $\delta(2t)$ has half area as $\delta(t)$

Also say $(2x) = u$; $2dx=du$

$$\int_{-\infty}^{+\infty} \delta(2x)dx = \frac{1}{2} \int_{-\infty}^{+\infty} \delta(u)du = \frac{1}{2} \int_{-\infty}^{+\infty} \delta(x)dx \text{ since } u \text{ is dummy variable;}$$

$$\text{So, } \delta(2t) = \frac{1}{2} \delta(t)$$

b) Based on the definition of the unit step and unit impulse functions, the relationship between the unit step and unit impulse functions is given by:

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Let $\tau = t - \sigma$ then: $u(t) = \int_0^\infty \delta(t - \sigma) d\sigma$. Hence proved.

c) The answer depends upon how we define $u_\Delta(t)$. $u_\Delta(t)$ in limiting form is known as a generalized function (as per the function is not specific). We can prove the given equality if we assume $u_\Delta(t)$ as t/Δ for $0 \leq t \leq \Delta$

$$\lim_{\Delta \rightarrow 0} [u_\Delta(t) \delta(t)] = \lim_{\Delta \rightarrow 0} [u_\Delta(0) \delta(t)] = \lim_{\Delta \rightarrow 0} [0] = 0$$

d)

$$\lim_{\Delta \rightarrow 0} [u_\Delta(t) \delta_\Delta(t)] = \lim_{\Delta \rightarrow 0} \left[\frac{t}{\Delta} \times \frac{l}{\Delta} \right] = \lim_{\Delta \rightarrow 0} \left[\frac{t}{\Delta^2} \right] = \frac{l}{2} \delta(t)$$

Note that as t is between 0 and Δ and as Δ is very small, and thus $\frac{\partial t}{\partial \Delta} \approx 1$.

$$\begin{aligned} \text{9) } y(t) &= \int_{-\infty}^t (\delta(\tau + 2) - \delta(\tau - 2)) d\tau = 0, \quad t \leq -2 \\ &= 1, \quad -2 < t \leq 2 \\ &= 0, \quad t > 2 \end{aligned}$$

$$\text{Thus, total energy} = E_\infty = \int_{-\infty}^{\infty} |y(t)|^2 dt = \int_{-2}^2 dt = 4$$